

2025 DHS Prelim Paper 3 Suggested Solutions

- 1 (a) gravitational field strength at a point is defined as the gravitational force exerted per unit mass placed at that point. **B1**

- (b) Since gravitational field strength g is the gravitational force F exerted per unit mass,

$$g = \frac{F}{m}$$

and the gravitational force exerted by the point mass on another body with mass m is $F = \frac{GMm}{r^2}$ **M1**

$$g = \frac{GMm}{mr^2} = \frac{GM}{r^2} \quad \textbf{A0}$$

Note: Do not accept answers with working but no explanations.

- (c) (at surface,) lines (of force) are radial **B1**

Earth has large radius / height above surface is small so lines are (approximately) parallel **B1**

and equally spaced **B1**